NIKE

How Nike Uses Customer Analytics to Drive Personalization

Customer Analytics: Spring 2025
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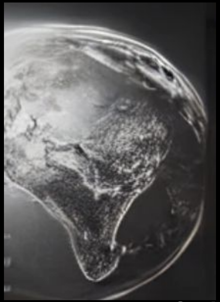
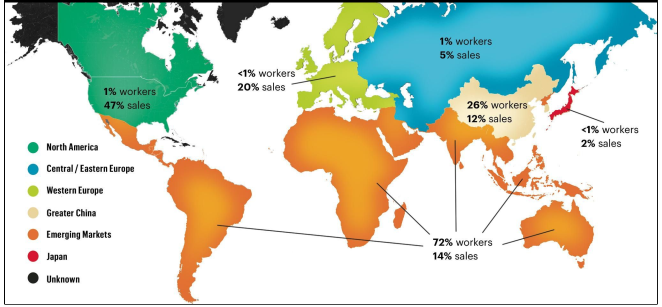
Agenda

- Company Overview
- Nike's Data Strategy
- Customer Journey Map
- Existing Applications of Customer Analytics
- AI Canvas
- Analytical Tools
- Application of R

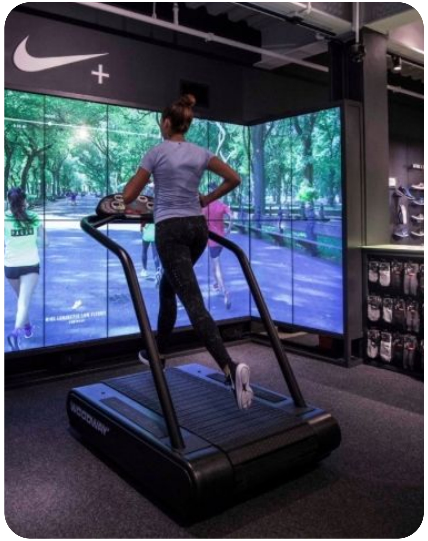


Global Presence



Company Overview

- Nike is one of the largest and best-recognized global sports and athletic wear brands.
- Headquartered near Beaverton, Oregon, and operates in over 190 countries worldwide. As of 2024, it employed 83,700 people worldwide.
- Largest sportswear company globally, with over $51 billion in annual revenue and a strong presence across retail and digital platforms.
- Invests heavily in technology and innovation, including fitness apps and smart wearables, to enhance consumer engagement.
- The “Just Do It” slogan focuses not on the glory of winning, but the hard work and daily struggle of putting in the effort no matter what.



Nike's Data Strategy

Data Collection & Integration

Where does Nike collect consumer data?

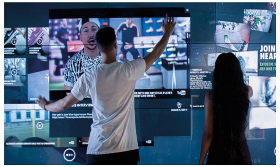
Nike Apps & Website: Nike gathers user engagement data through web tracking tools, APIs, and SDKs on Nike.com, SNKRS, Nike Training Club, and Nike Run Club apps

E-commerce & Point Of Sale Systems: Nike collects transactional data through online and in-store purchases and store real-time data in POS

Wearables & Fitness Integrations: Nike apps integrate seamlessly with Apple Health, Google Fit, and Strava to capture running data and workouts

Social Media & Digital Engagement: Nike analyzes customer interactions and feedback using sentiment analysis and NLP on Instagram, TikTok, and other social media

IoT & RFID Technology: Nike uses RFID-tagged products in stores and connected apparel (e.g., smart shoes) to track user engagement



Data Ingestion & Processing

What does Nike use to process consumer data?

Cloud Storage & Computing (AWS, Azure): Nike utilizes cloud data lakes and data warehouses to store and process structured and unstructured consumer data

Customer Data Platform: Nike uses master data management and identity resolution and user matching to integrate data from multiple sources into a unified view, showing all relevant data under individual customer profiles

ETL Pipelines: Nike sets up ETL pipelines that cleanse, format, and standardize raw data before analysis

Machine Learning & AI Models: Nike applies AI-driven insights to detect patterns, predict behavior, and automate personalization

Apache Kafka & Google Pub/Sub: Nike uses these platforms to ingest real-time data across the digital ecosystem

Nike's Data Strategy

Data Activation

How does Nike analyze data to capture insights?

Recommendation Engines: Nike uses AI-powered algorithms to suggest products based on browsing activity, purchasing history, and workout behavior

Dynamic Pricing & Offers: Nike develops real-time pricing models based on demand and past consumer behavior

Contextual Messaging & Notifications: Nike uses data to cluster customers into segments to serve targeted marketing, loyalty rewards, and relevant product launches

Retail Personalization: Nike leverages omnichannel integration that allows in-store staff to pull up customer preferences and suggest relevant items

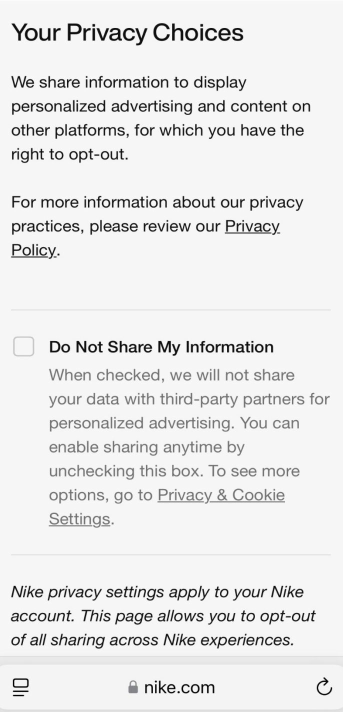
Data Governance & Privacy Compliance

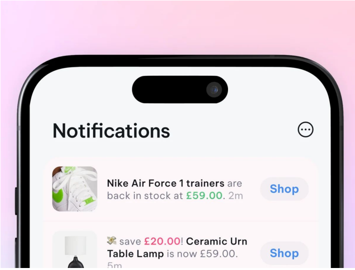
How does Nike protect data throughout its lifecycle?

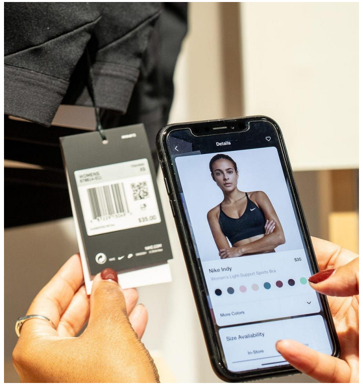
Encryption & Anonymization: Nike implements end-to-end encryption for consumer transactions and stored data and anonymizes personal information for when training AI algorithms

User Consent Management: Nike informs consumers when their data is being collected and allows users to opt out of data sharing through its website and apps

Fraud Detection & Risk Analysis: Nike uses AI models to detect anomalies in transactions to prevent fraud in sneaker drops and resale markets





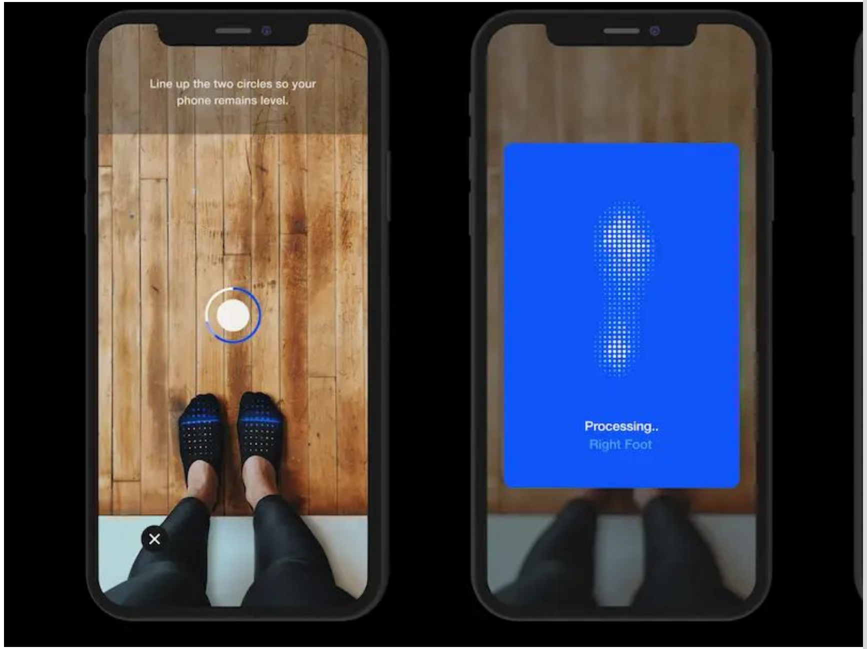


Customer Journey Map

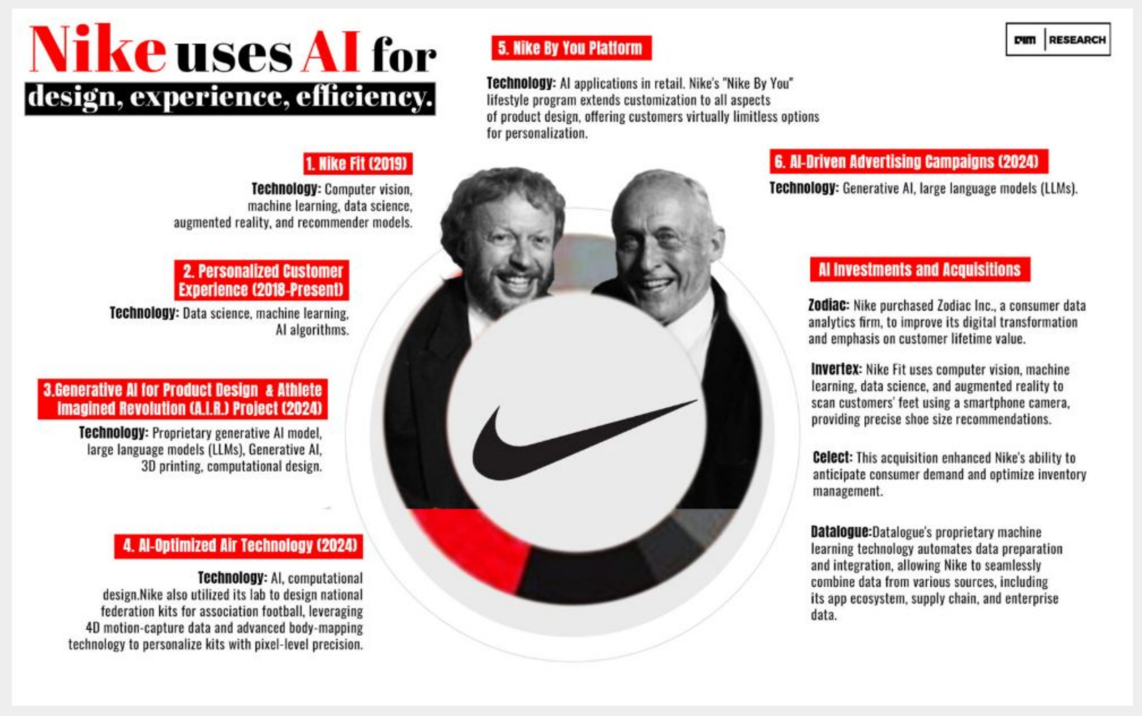
	Awareness	Consideration	Purchase	Usage	Loyalty
Touchpoint	Nike.com Nike App Nike Physical Stores Athlete Endorsements	Product Pages Customization Options Nike App Social Media Engagement	Online Shopping In-Store Purchases Payment Options Nike By You	Returns and Exchanges Nike App Customer Support NikePlus Membership	Exclusive Drops Personalized Offers Newsletter Feedback Channels
Customer Activities	Customer visits website Customer downloads and engages with Nike app Customer encounters Nike's physical stores Customers are influenced by athlete endorsements	Customer explores product pages on Nike.com Customer uses Nike By You to design shoes Customer browses Nike App Customer engages with Nike social media content	Customer makes online purchase through Nike.com or physical retail stores Customer chooses from various payment options Customer finalizes nad purchases personalized designs	Customer returns/exchanges products Customer uses Nike App to track workouts Customer contacts customer support Customer uses NikePlus benefits	Customer gets early access to limited-edition product releases Customer receives tailored promotions and discounts Customer subscribes to receive exclusive news and updates Customer provides feedback
Data Collected	User preferences, purchase history, workout habits, content engagement, keyword searches, account creation data (demographics)	Product preferences, browsing history, time spent on site, clickstream data, abandoned cart behavior, RFID & beacon technology	Transactional data (purchase history, payment info, order value, discount/promo usage), shipping, inventory & fulfilment	App usage patterns, return & refund data, post-purchase surveys & reviews, chatbot & customer service interactions	Email address for newsletter, open rates/click through rates, loyalty points & rewards, membership activity, bundling & upselling patterns
Data Formats	Social Media Data, XML (product catalog and API integrations)	Log Files (clickstream, checkout behavior)	CSV (stores transactional records, purchase history, inventory data)	JSON (Nike App interactions, customer profiles), Text Data (Customer reviews, chatbot transcripts)	SQL Databases (loyalty programs)

Nike's Customer Analytics

1. **<u>Nike Fit App</u>**
    a. **Foot Measurement Technology:** Uses AI and computer vision to scan and measure customers' feet for personalized shoe size recommendations.
    b. **Data Utilized:** Foot shape, arch height, width, length and fit preferences.
    c. **Outcome:** Accurate sizing recommendations which improves product design and customer experience.
2. **<u>Nike Training Club and Nike+ App</u>**
    a. **Customer Feedback Loop:** Analyzes customer reviews and feedback for product improvement
    b. **Wearable Tech Insights:** Collects data from products like Nike+ App and Nike Training Club to understand customer exercise patterns and preferences.
    c. **Outcome:** Guides product improvements and new product creation based on real-time customer data.
3. **<u>Predictive Analytics</u>**
    a. **Demand Forecasting:** Uses historical sales and customer data to forecast demand.
    b. **Outcome:** reduces overstocking and understocking, ensuring products are available when customers want them.



Nike's AI Journey



Nike leverages AI and Gen AI to **enhance consumer experience, streamline supply chain, and drive product innovation.**

Nike actively engages in customer feedback to drive product improvements

1 Continuous Improvement Based on Customer Insights

To deliver exceptional customer experience. For eg., after feedback on checkout complexity, they streamlined the process, boosting conversion rates by **15%**, showcasing their commitment to exceptional customer experience.

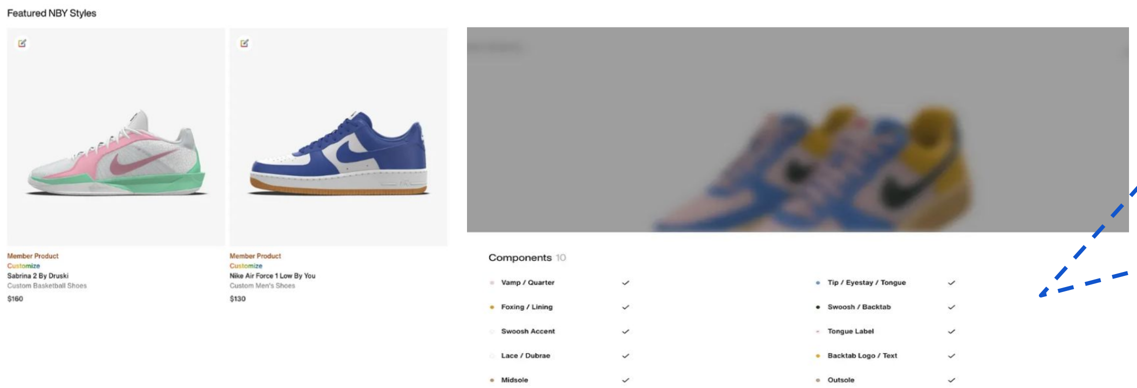
2 Incorporating Feedback into Product Development

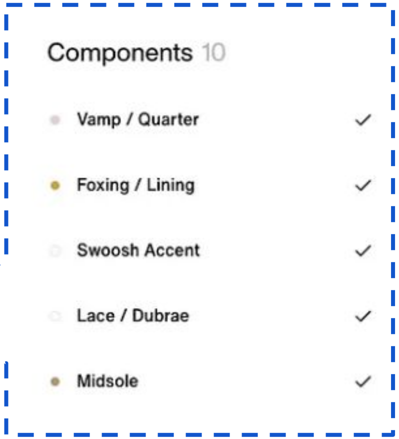
To meet customer needs. The Nike React running shoe, for eg., was designed based on runners' input for a lightweight, durable shoe, leading to a successful, best-selling product.

3 Listening to Customers Through Digital Channels

For identifying trends and areas of improvement. For eg., feedback from Nike App led to personalized homepages and product recommendations, boosting user satisfaction and engagement.

Nike By You: Leveraging Customer Data for Personalization





Nike empowers users to **'make their own'** unique product - offering the company insights into **consumer preferences, behaviour, and trends** which allows Nike to leverage it for: Product Development, Marketing Strategies and improving customer loyalty

Nike's investment in predictive customer analytics drives DTC sales, making it a key driver of future growth

ZODIAC

Acquired in 2018

Nike acquired Zodiac, a marketing analytics firm specializing in customer habit analysis and purchase prediction which enabled it to:

- **Digitize** it's operations via the Fit App integration,
- **Personalize** in-app product recommendations and content delivery

celect

Acquired in 2019

Nike acquired a demand-sensing analytics platform that provides insights for inventory optimization across various channels. By integrating Celect's technology, Nike can

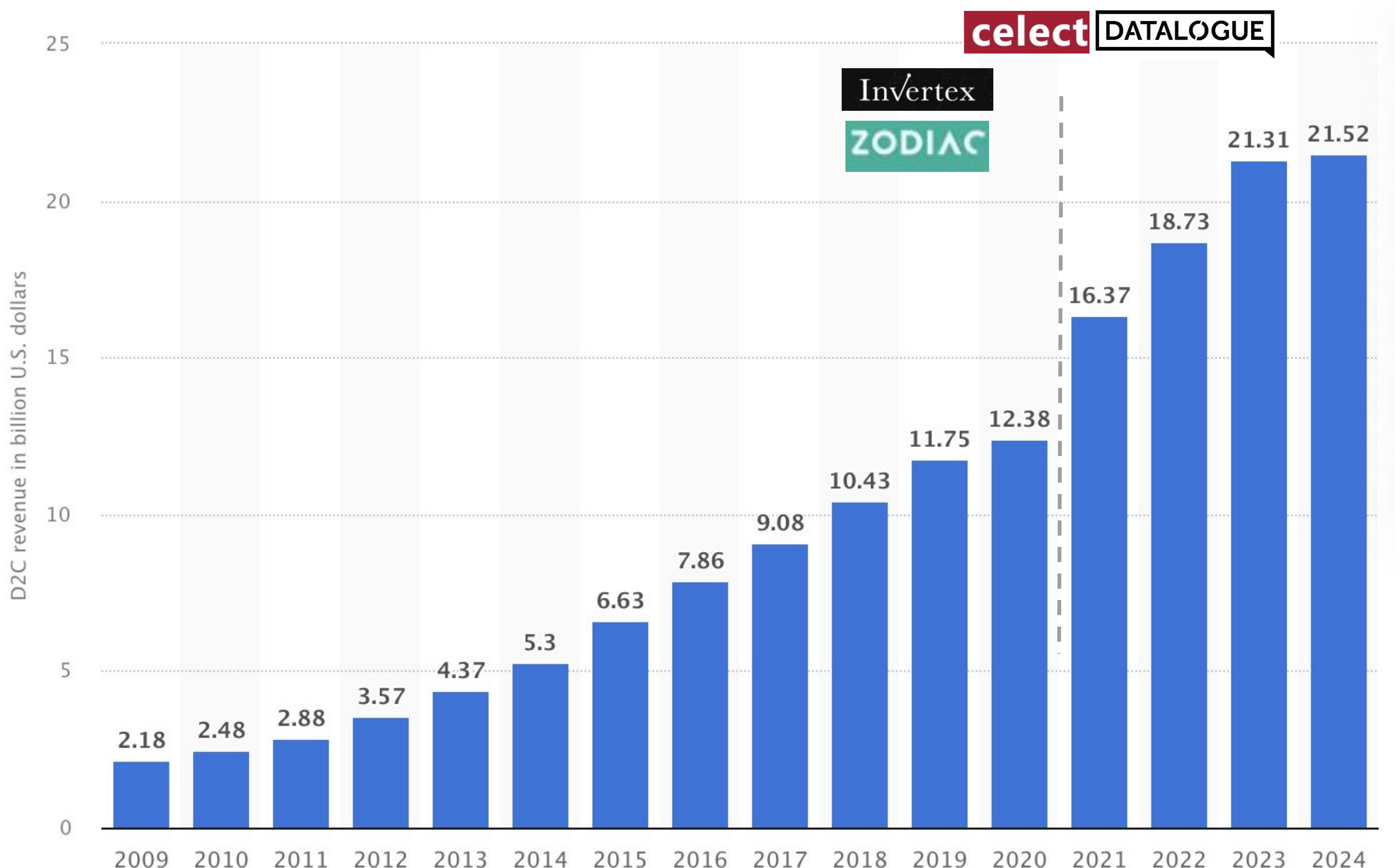
- Make **informed decisions** about product manufacturing and distribution,
- Ensure that inventory **aligns with localized demand.**

+12% Direct Sales in one year

+32% Digital Sales in one year

Note: The figures are a result of the overall integration of customer analytics tools with Nike's AI and machine learning technology in its digital ecosystem

Key technology acquisitions and investments in cloud-native microservices have contributed to 73% increase in Nike's DTC revenue in the last 5 years



AI Canvas

Predicting how Nike is leveraging AI-driven analytics to enhance customer experience, optimize inventory management, and drive digital sales.

Prediction

Nike's AI predicts customer preferences, demand trends, and optimal inventory levels based on historical sales data, user engagement, and real-time customer interactions.

Judgment

If right → Customers receive highly personalized product recommendations, better sizing suggestions, and enhanced shopping experiences, leading to increased satisfaction and sales.

If wrong → Poor recommendations or inventory mismanagement can lead to customer dissatisfaction, higher return rates, and financial losses.

Action

- **Nike Fit App**: Uses AI and computer vision for precise shoe size recommendations.
- **Nike Training Club & Nike+ App**: Leverages AI to analyze workout data and enhance product recommendations.
- **Predictive Analytics**: Implements demand forecasting to optimize stock levels and prevent overstocking or understocking.

Outcome

- Enhanced customer experience and satisfaction
- Increased customer loyalty and sales
- Reduced inventory costs and improved supply chain efficiency

Training

- Customer purchase history and browsing behavior
- Product attributes and fit data
- Engagement with digital platforms (Nike+ App, Training Club, website interactions)
- Historical sales and inventory trends

Input

- Customer demographics and behavioral data
- Product interaction history (views, add-to-cart, purchases, returns)
- Real-time demand and inventory data

Feedback

- Customer reviews and product feedback
- Return and refund data
- Engagement metrics from digital tools and personalized recommendations

How will this AI impact on the overall workforce

- Strengthened customer engagement and brand loyalty
- More accurate demand forecasting, reducing waste and improving efficiency
- Enhanced digital shopping experience with better recommendations and sizing
- Employees may require training to effectively work with AI-driven tools
- Evolution of new roles focusing on AI integration and data analysis

Applications of R - ***recommenderlab***

provides a framework for developing and testing recommendation algorithms, supporting collaborative filtering and content-based methods.

Key Features

- Supports user-based and item-based collaborative filtering
- Provides evaluation metrics for recommendation performance
- Handles various types of recommendation data structures

How to Utilize the Package

- Load the package and a dataset (e.g., MovieLense dataset).
- Convert data into a realRatingMatrix.
- Split data into training and test sets using evaluationScheme().
- Train a recommendation model using Recommender().
- Generate and evaluate predictions using predict() and calcPredictionAccuracy().

Nike Application

Nike could use recommenderlab to develop personalized product recommendations for customers based on their purchase history, browsing behavior, and interactions on the Nike+ app.

1. **Load the data in R:**

```r
library(recommenderlab)
data <- read.csv("user_item_matrix.csv", row.names = 1)
```

2. **Convert to a `realRatingMatrix`:**

```r
rating_matrix <- as(as.matrix(data), "realRatingMatrix")
```

3. **Train a recommendation model (e.g., Item-Based Collaborative Filtering):**

```r
recommender <- Recommender(rating_matrix, method = "IBCF")
```

4. **Generate predictions:**

```r
predictions <- predict(recommender, rating_matrix[1, ], n = 5)
as(predictions, "list")
```

Applications of R - ***rrecsys***

provides implementations of various recommendation algorithms, better for optimizing and fine-tune a scalable production-ready model.

Key Features

- Includes multiple recommendation algorithms
- Provides matrix factorization techniques
- Allows flexible model evaluation

How to Utilize the Package

- Load data (e.g., mlLatest100k dataset).
- Create a recommendation model with Recommender() using a chosen algorithm (e.g., FunkSVD).
- Generate top-N recommendations using predict().

Nike Application

Nike can use rrecsys to fine-tune recommendation models by analyzing user reviews, product ratings, and interactions with Nike's mobile apps, improving the personalization of product suggestions.

Step 1: Install and Load `rrecsys`

Step 2: Read the User-Item Matrix

Step 3: Convert to `rrecsys` Data Format

The `rrecsys` package requires a **recommendation dataset** format:

```r
rec_data <- defineData(data_matrix)
```

Step 4: Train a Recommendation Model

A. Item-Based KNN Model

```r
model_ibknn <- rrecsys(rec_data, "ItemKNN")
```

Generate recommendations for the first user:

```r
predictions <- topN(model_ibknn, 1, n = 5)
print(predictions)
```

B. FunkSVD Model (Matrix Factorization)

```r
model_svd <- rrecsys(rec_data, "FunkSVD", k = 10)
predictions_svd <- topN(model_svd, 1, n = 5)
print(predictions_svd)
```

Sources

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Questions?